

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-090

WASTE DISCHARGE REQUIREMENTS
FOR
GOLDEN VALLEY UNIFIED SCHOOL DISTRICT
LIBERTY HIGH SCHOOL WASTEWATER TREATMENT FACILITY
MADERA COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Golden Valley Unified School District (hereinafter Discharger) submitted a Report of Waste Discharge (RWD), dated 22 May 2000, in support of discharge to land and reclamation of effluent from a new wastewater treatment facility (WWTF) to serve the proposed Liberty High School, northwest of the intersection of Avenue 12 and Road 36, Madera County. The school will be occupied 190 days/year. Its student and staff population will be 675 for Phase I and 1350 for Phase II for 190 days/year. The RWD includes a technical support document dated May 2000. The WWTF, 1.35-acre pond, and 5-acre use area are on land owned by the Discharger (Assessor's Parcel No. 34-226-05/06) in Section 33, T11S, R19E, MDB&M, as shown in Attachment A, which is part of this Order by reference.

Facility Description

2. The proposed WWTF will be built in two phases and consists of a package activated sludge plant with Phases I and II capacities of 20,000 and 40,000 gallons/day respectively. The WWTF includes flow monitoring, solids grinding, flow equalization, oxidation, clarification, digestion, filtration, and disinfection. The RWD states that the effluent will meet requirements for disinfected tertiary recycled water pursuant to Title 22, California Code of Regulations, Division 4, Chapter 3 (hereafter Title 22). Effluent disposal is by spray irrigation of turf at agronomic rates. During the fall and winter, effluent not required for irrigation will be stored in an unlined 7.9 acre-foot capacity pond with a surface area of 1.35 acres. The treatment schematic and flow diagram are shown in Attachment B, which is part of this Order by reference.
3. The Discharger will hold waste activated sludge in a sludge storage tank and periodically truck the sludge to a publicly owned treatment works for further treatment and disposal.
4. The RWD states that the use area will be an area outside of the school facility. The Discharger plans to fence the pond area to preclude direct public access.
5. The RWD states that there will be a standby emergency power generator of sufficient size to operate the treatment plant and ancillary equipment during periods of loss of commercial power.

Hydrology, Soils and Land Use

6. The proposed WWTF and use area are within the San Joaquin Valley Hydrologic Basin, San Joaquin Valley Floor Hydrologic Unit, Berenda Creek Hydrologic Area (No. 545.30) as depicted on interagency hydrologic maps prepared by the California Department of Water Resources (DWR) in August 1986. The slope is to the southwest with a grade of less than 0.1%. The site is approximately one mile north of Root Creek and four miles north of the San Joaquin River.
7. Site soils are predominantly loamy sand. The RWD predicts stored effluent will percolate at a rate of 0.25 in/day. This rate will yield an annual percolation of 7.6 feet or 10.3 acre-feet/year. At an estimated effluent nitrate-nitrogen concentration of 45 mg/L, this is equivalent to an annual nitrogen loading of 933 lbs-N/acre/year.
8. The average precipitation in the area is 10.9 inches, according to data from the California Irrigation Management Irrigation System (CIMIS) weather station for the period of 1965 through 2000.
9. The water balance in the RWD utilizes a 100-year return frequency annual precipitation for the project vicinity of 21.6 inches and a wet year evaporation of 60 inches.
10. Areal groundwater is approximately 140 to 160 feet below ground surface. Although the regional groundwater flows southwesterly, there is a localized depression such that groundwater flow is to the northwest at the project site according to information in *Lines of Equal Elevation of Water in Wells in Unconfined Aquifers*, published by DWR in Spring 1997. The RWD indicates that well yields in the project vicinity are typically 500-600 gallons/minute.
11. According to the RWD, Liberty High School will obtain domestic and irrigation water from on-site wells.
12. The existing land use to the north, west, and south of the project is agricultural. The land to the east of the project is residential.

Basin Plan and Beneficial Uses

13. The Board adopted a *Water Quality Control Plan for the Sacramento River Basin and the San Joaquin River Basin, Fourth Edition* (hereafter Basin Plan), which designates beneficial uses, establishes water quality objectives, and contains implementation plans and policies for all waters of the basins. These requirements implement the Basin Plan.
14. The Basin Plan identifies the present and anticipated beneficial uses of underlying groundwater as domestic, industrial, and agricultural supply.

15. Nitrogen requirements for turf range from about 90 to 220 pounds/acre/year, according to *A Guide to Turfgrass Nutrient Recommendations*, Kansas State University and Agricultural Experiment State and Cooperative Extension Service, MF-2311, January 1998.
16. The Discharger has estimated the effluent nitrogen concentration at 45-50 mg/L. At a total annual flow of 7.6 million gallons, this is approximately 3,170 lbs N/year.

Water Recycling Criteria

17. The California Department of Health Services (DHS) has established statewide criteria and adopted revisions to the water recycling criteria contained in Title 22 on 2 December 2000. The revised Title 22 requires that all wastewater used for reclamation on school grounds be a disinfected tertiary recycled water.
18. The 1996 Memorandum of Agreement (MOA) between DHS and the State Water Resources Control Board on the use of recycled water establishes basic principles relative to the agencies and the regional boards. In addition, the MOA allocates primary areas of responsibility and authority between these agencies, and provides for methods and mechanisms necessary to assure ongoing, continuous coordination of activities relative to the use of recycled water in California.
19. Section 60323, Title 22, dated 2 December 2000, requires recyclers of treated domestic wastewater to submit an engineering report detailing the use of recycled water, contingency plans, and safeguards. The Discharger has filed a Title 22 engineering report. By letter dated 2 June 2000, the DHS has approved the project with recommendations for a continuous chlorine analyzer/recorder and alarm system.
20. By letter dated 7 March 2001, DHS requested: (a) Discharge Specification B.3.b. be reworded to comply with Title 22, Section 60301.230(b); and (b) including the specification that the filtration rate does not exceed 5 gpm/ft² as specified in Title 22, Section 60301.320(a)(1).

Antidegradation and CEQA

21. The Facility described in Finding No. 2 provides best practicable treatment or control for the subject wastewater intended to assure that the discharge does not create a condition of pollution or of nuisance and that the highest water quality defined by groundwater limitations will be maintained. Further, the discharge is consistent with the Board's reclamation policy and complies with the requirements for the use of recycled water for irrigation of school yards as set forth in Title 22.
 - a. The treatment process incorporates:
 - i. technology for coagulated, filtered secondary treatment of domestic wastewater,
 - ii. biosolids handling and disposal,

- iii. disinfection process to facilitate wastewater recycling,
 - iv. recycling of wastewater at agronomic rates.
- b. Control measures include:
- i. sufficient staffing to assure proper operation and maintenance,
 - ii. standby emergency power generator of sufficient size to operate the treatment plant and ancillary equipment during periods of loss of commercial power.
22. The conditional discharge as permitted herein is consistent with the antidegradation provisions of State Water Resources Control Board Resolution No. 68-16. The depth to groundwater ranges from 140 to 160 feet. As such, it is reasonable to assume that there is assimilative capacity in the soils and underlying groundwater, considering limited dilution, to allow for some degradation and not unreasonably threaten present and anticipated beneficial use of such water or result in concentrations of waste constituents in groundwater that exceeds or threatens to exceed water quality objectives set forth in the Basin Plan. Such degradation, if it occurs, will likely be limited to groundwater underlying the pond. Some degradation of groundwater immediately underlying the effluent storage is allowable provided the discharge reflects the implementation of BPTC measures and the degradation is (a) consistent with the maximum benefit to the people of the State, (b) will not unreasonably affect present and anticipated beneficial use of affected groundwater, and (c) will not exceed water quality objectives.
23. On 11 July 2000, the Golden Valley Unified School District Board of Trustees certified the project's Final Environmental Impact Report, adopted Statements and Findings of Fact, adopted a Statement of Overriding Considerations, and adopted a Mitigation Monitoring Plan in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000 et seq.) and State CEQA guidelines. The Board considered the Environmental Impact Report and associated CEQA documents and concurs that the project will have no significant effect on water quality.

General Findings

24. Authorization to discharge is conditional upon the discharge complying with provisions of Division 7 of the California Water Code (CWC) and any more stringent effluent limitations necessary to implement water quality control plans, to protect beneficial uses, and to prevent nuisance. Compliance with this Order should assume this and mitigate any potential adverse changes in water quality due to discharge. Pursuant to CWC Section 13263(g), discharge is a privilege, not a right and issuance of this Order does not create a vested right to continue the discharge. Failure to provide the level of management required to assure BPTC, preclude

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conditions that threaten pollution or nuisance, and maintain the highest quality groundwater will be sufficient reason to modify this Order, revoke the Order, or enforce the Order and prohibit further discharge.

25. State regulations pertaining to water quality monitoring for waste management units are found in Title 27, CCR, Section 20380 et seq., (hereafter Title 27). These regulations prescribe procedures for detecting and characterizing the impact of waste constituents on groundwater. While the facility has been found exempt from Title 27, the data analysis methods of Title 27 are appropriate for determining whether the discharge complies with the terms for protection of groundwater specified in this Order.
26. The Discharger must obtain coverage under an NPDES general industrial storm water permit. In order to obtain coverage of the general industrial permit, Discharger must submit a Notice of Intent (NOI) to comply with the permit to the State Water Resources Control Board and must prepare a Storm Water Pollution Prevention Plan (SWPPP) for the operation of the facility.
27. Because construction associated with the project will disturb more than five acres, Discharger must comply with the NPDES General Permit No. CAS000002 for Discharges of Storm Water Associated With Construction Activity. Before construction begins, the Discharger must also submit an NOI to comply with the General Construction Permit to the State Water Resources Control Board and must prepare a SWPPP for the construction of the facility.
28. The Board has considered the supplemental data and information in the Information Sheet in making findings, and adopting terms and conditions of this Order, and attaches it as part of this Order as documentation.
29. The Board notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
30. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Golden Valley Unified School District, its agents, successors and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following at Liberty High School Wastewater Treatment Facility:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. The bypass or overflow of untreated or partially treated waste is prohibited, except as allowed in Provision E.2 of Standard Provisions and Reporting Requirements.

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3. Discharge of wastes to land other than the use area and the pond described herein is prohibited.
4. Discharge of waste classified as 'hazardous,' as defined in Section 2521(a) of Title 23, CCR, Section 2510 et seq., or 'designated,' as defined in Section 13173 of the California Water Code, is prohibited.

B. Discharge Specifications

1. **After Provision F.3 is satisfied**, the Phase I maximum daily discharge flow shall not exceed 20,000 gallons/day.
2. **After Provision F. 4 is satisfied**, the Phase II maximum daily discharge flow shall not exceed 40,000 gallons/day and the total annual discharge flow shall not exceed 7.6 million gallons.
3. Discharge to the pond shall not exceed the following limits:
 - a. Five day, 20°C biochemical oxygen demand (BOD₅):
 - i. 10 mg/L 30-day average.
 - ii. 30 mg/L daily maximum.
 - b. **Total Coliform.** The median concentration of total coliform bacteria shall not exceed a most probable number (MPN) of 2.2/100 mL utilizing the bacteriological results of the last seven days for which analyses have been completed and the number of total coliform bacteria shall not exceed 23 MPN/100 mL in more than one sample in any 30 day period. No sample shall exceed an MPN of 240 coliform bacteria per 100 mL.
 - c. **Turbidity.** The turbidity shall not exceed any of the following:
 - i. A daily average of 2 Nephelometric Turbidity Units (NTU);
 - ii. 5 NTU more than 5 percent of the time within a 24-hour period; **and**
 - iii. 10 NTU at any time.
 - d. **Chlorine Residual.** The product of total chlorine residual and modal contact time shall not be less than 450 milligram-minutes/liter at all times with a modal contact time of at least 90 minutes, based on peak dry weather design flow. The total chlorine residual concentration shall not be less than 5 mg/L.

- e. **Electrical Conductivity (EC).** The maximum EC of the discharge shall not exceed the average EC of the source water plus 500 $\mu\text{mhos/cm}$, or a total of 1,000 $\mu\text{mhos/cm}$, whichever is less.
- 4. The filtration rate shall not exceed 5 gallons per minute per square foot of surface area in mono, dual or mixed media gravity, upflow or pressure filtration systems.
- 5. The WWTF, pond, and use area shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
- 6. The WWTF shall meet the reliability criteria set forth in Title 22, Sections 60345, 60347, 60353, and 60355.
- 7. Objectionable odors originating at the WWTF shall not be perceivable beyond the limits of the wastewater treatment and use area.
- 8. As a means of discerning compliance with Discharge Specification B.6, the dissolved oxygen content in the upper zone (1 foot) of the pond shall not be less than 1.0 mg/L.
- 9. The pond shall be managed to prevent breeding of mosquitoes. In particular:
 - a. An erosion control plan should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth and harvesting.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
 - d. Vegetation management operations in areas that attract nesting birds shall be carried out either before or after, but **not during**, the **April 1 to June 30** bird nesting season.
- 10. The pond shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet in any pond (measured vertically). A permanent marker shall be installed in the pond.
- 11. Reclamation of WWTF effluent shall be at reasonable agronomic rates considering the crop, soil, climate, and irrigation management plan. The annual nutrient loading of reclamation areas, including the nutritive value of organic and chemical fertilizers and of the recycled water shall not exceed the crop demand.

12. On or about **1 October** of each year, available pond capacity shall be at least equal to the volume necessary to comply with Discharge Specification B.11.

C. Recycled Water Specifications

1. Public contact with wastewater in the treatment and use area shall be precluded as follows:
 - a. A six-foot high chain link fence shall surround the treatment plant and pond.
 - b. All piping, valves, and outlets shall be marked to differentiate recycled water from domestic or other water. All reclamation water lines shall be constructed of purple pipe to indicate nonpotable water.
 - c. All recycled water components such as controllers shall be affixed with recycled water warning signs.
 - d. The use area shall be posted with signs that are visible to the public, in a size no less than 4-inches high by 8-inches wide, that include the following wording:

“RECYCLED WATER – DO NOT DRINK.”

“AGUA DE DESPERDICIO RECLAMADA—POR FAVOR NO TOME”

Each sign shall display an international symbol similar to that shown in Attachment C.

- e. No hose bibbs shall be installed on the recycled water distribution system.
 - f. All valves, outlets, quick couplers, and sprinkler heads serving the recycled water system shall be of a type that can only be operated by an authorized user.
 - g. Recycled water irrigation shall be conducted during the hours when the turf area is unoccupied and the recycled irrigation system shall be operated to prevent wind drift of the spray from the use area.
2. Recycled water shall be managed to minimize contact with workers.
3. The Discharger shall maintain the following setback distances from the pond and areas irrigated with recycled water:

<u>Setback Distance (feet)</u>	<u>To</u>
100	Residences/Irrigation Wells
100	Domestic Wells

4. Areas irrigated with recycled water shall be managed to prevent breeding of mosquitoes. More specifically:
 - a. All applied irrigation water must infiltrate completely within a 48-hour period.
5. Low-pressure and unpressurized pipelines and ditches accessible to mosquitoes shall not be used to store recycled water.

D. Sludge Disposal Specifications:

1. Collected screenings, grit, sludge, and other residual sludges and solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer, and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005 et seq.
2. Any proposed change in sludge disposal practice shall be reported to the Executive Officer **at least 90 days in advance** of the change.

E. Groundwater Limitations

The WWTF operation and its discharges to land shall not cause degradation of area groundwater in excess of that resulting from the implementation of the best practicable treatment and control identified herein, or violate Standard Provision A.11.

F. Provisions

1. The Discharger shall comply with Monitoring and Reporting Program (MRP) No. 5-01-090 which is a part of this Order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with *Standard Provisions and Reporting Requirements*, dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as Standard Provision(s).
3. By **30 days prior** to Phase I WWTF start-up, the Discharger shall submit certification that the proposed WWTF, pond, and use area have been built as designed and are capable of complying with the terms of this Order. The certification report shall be prepared and certified by a California registered civil engineer with experience in wastewater treatment and disposal facility design and shall include a map depicting features such as the WWTF, pond, use area, and the location of signs described in Recycled Water Specifications C.1.d. Upon written approval by the Executive Officer, this Provision will be considered satisfied.
4. By **30 days prior** to Phase II WWTF startup, the Discharger shall submit certification that the proposed WWTF expansion has been built as designed and is capable of complying with

the terms of this Order. The certification report shall be prepared and certified by a California registered civil engineer with experience in wastewater treatment and disposal facility design. Upon written approval by the Executive Officer, this Provision will be considered satisfied.

5. By **30 days prior** to Phase II WWTF startup, the Discharger shall submit an irrigation management plan to assure that nitrogen application is within the agronomic rates for the crop thereby minimizing the percolation of nitrogen to groundwater. The certification report shall be prepared and certified by a California registered civil engineer with experience in wastewater treatment and disposal facility design. Upon written approval by the Executive Officer, this Provision will be considered satisfied.
6. The Discharger shall maintain a standby emergency power generator of sufficient size to operate the WWTF and ancillary equipment during periods of loss of commercial power. The standby emergency power generator shall start and stop automatically and power transfer shall occur automatically.
7. The Discharger shall not allow pollutant-free wastewater to be discharged into the collection, treatment, and disposal system in amounts that significantly diminish the WWTF's capability to comply with this Order. Pollutant-free wastewater means rainfall, groundwater, cooling waters, and condensates that, without treatment, are essentially free of pollutants.
8. The Discharger shall implement best practicable treatment and control, including proper operation and maintenance, to comply with this Order.
9. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
10. If there is evidence that reasonable potential exists that pond percolation is contributing to degradation of groundwater passing beneath and downgradient of the WWTF in violation of the State Antidegradation Policy, Resolution 68-16, the Board may reopen this Order for consideration of adopting numerical limitations for groundwater passing beneath and downgradient of the WWTF vicinity that are consistent with maximum benefit to the people of the State, will not unreasonably affect present and anticipated beneficial use of affected groundwater, and will not exceed water quality objectives.
11. By **60 days prior** to WWTF start-up, the Discharger shall submit a list of the names, addresses, and telephone numbers of contact persons such as the WWTF operator and school principal.
12. By **60 days after** WWTF start-up, the Discharger shall submit a complete copy of the WWTF's operations and maintenance manual.

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13. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.
14. A copy of this Order shall be kept at the WWTF for reference by personnel responsible for the maintenance and operation of the WWTF. Key operating personnel shall be familiar with its contents.
15. In the event of any change in control or ownership of the WWTF presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the state of incorporation if a corporation, the names and address and telephone number of the persons responsible for contact with the Board, and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved by the Executive Officer.

16. Within **30 days** following any change in name or title of contact persons (e.g., such as chief administrative officer or chief plant operator), the Discharger shall notify the Board in writing of such change and identify the name(s), title(s), address(es), and telephone number(s) of such person(s).
17. The Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 27 April 2001.

GARY M. CARLTON, Executive Officer

BLH:fmc: 4/27/01

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-090

MONITORING AND REPORTING PROGRAM
FOR
GOLDEN VALLEY UNIFIED SCHOOL DISTRICT
LIBERTY HIGH SCHOOL WASTEWATER TREATMENT FACILITY
MADERA COUNTY

Specific sample station locations shall be established at the wastewater treatment facility (WWTF) with concurrence of the Board's staff. A description of the stations shall be submitted to the Board and attached to this Program.

INFLUENT MONITORING

Influent samples shall be collected at the inlet of the headworks and at approximately the same time as effluent samples. Influent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
BOD ₅ ¹	mg/L	8-hr composite ²	2/month ³
Total Suspended Solids	mg/L	8-hr composite ²	2/month ³
Settleable Solids	mL/L	Grab	2/month ³

¹ Five-day, 20°C biochemical oxygen demand

² Composite samples may consist of flow-proportioned grab samples.

³ During nonconsecutive weeks

EFFLUENT MONITORING

Flow may be measured either at the headworks or at the outlet from the treatment system (i.e., discharge to the pond). Effluent samples shall be collected subsequent to disinfection and prior to discharge to the pond. The sampling frequency shall be as indicated below unless results of monitoring a pollutant appear to violate effluent limitations, but monitoring frequency is not sufficient to validate violation (e.g., the monthly average for BOD₅) or indicate a violation and potential upset of the treatment process (e.g., greater than maximum coliform). In such cases, the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem. Effluent samples should be representative of the volume and nature of the discharge. The time of collection of a grab sample shall be recorded. The following shall comprise the effluent monitoring program for the WWTF:

<u>Constituents</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sampling Frequency</u>
Total Daily Flow	gallons/day	Metered	Continuous
Chlorine Residual	mg/L	Metered	Continuous
Turbidity	NTU	Metered	Continuous
BOD ₅	mg/L	8-hour composite ¹	Weekly
Total Suspended Solids	mg/L	8-hour composite ¹	Weekly

<u>Constituents</u>	<u>Units</u>	<u>Sample Type</u>	<u>Sampling Frequency</u>
Total Kjeldahl Nitrogen	mg/L	8-hour composite ¹	Monthly
Nitrate-Nitrogen	mg/L	8-hour composite ¹	Monthly
Total Coliform Organisms	MPN ² /100 mL	Grab ³	Daily
EC ⁴	µmhos/cm	Grab	Weekly
pH	pH units	Grab	Weekly
<u>Minerals</u> ⁵	mg/L	Grab	Annual ⁶

¹ Composite samples may consist of flow-proportioned grab samples.

² Most Probable Number.

³ Total coliform organisms samples shall be dechlorinated immediately following sample collection as part of sample collection procedures.

⁴ Conductivity at 25 °C.

⁵ Minerals, as referenced in this program, shall include: aluminum, alkalinity, bicarbonate, boron, calcium, carbonate, chloride, fluoride, iron, magnesium, manganese, total phosphorous, potassium, sodium, sulfate, and total dissolved solids. An anion cation balance shall accompany the analyses to demonstrate that the analyses are complete. TDS and inorganic fractions of TDS shall be determined using EPA Test Method Nos. 160.1 and 160.4 respectively.

⁶ Collected during October.

POND MONITORING

The pond shall be monitored as follows:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
pH	pH units	Grab	Weekly
Freeboard	Feet	Observation	Weekly
<u>Dissolved Oxygen (DO)</u> ¹	mg/L	Grab	Weekly

¹ Samples shall be collected between 0800 and 0900 hours near the pond outlet and analyzed for dissolved oxygen. Samples shall be collected at a depth of one foot, opposite the inlet, and analyzed for D.O. If D.O. results indicate noncompliance with Discharge Specification B.8, the Discharger shall implement corrective measures as specified in the O&M manual and monitor the pond daily until its D.O. stabilizes above 1.0 mg/L.

The Discharger shall inspect the condition of the pond at a minimum of once per week and write visual observations in a bound logbook. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether burrowing animals or insects are present; and the color of the pond (e.g., dark sparkling green, dull green, yellow, gray, tan, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

WATER SUPPLY MONITORING

A sampling station (or series of stations) shall be established where a representative sample can be obtained of Liberty High School's water supply. Water supply monitoring shall be conducted concurrent with effluent monitoring and include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Sampling Frequency</u>
EC	µmhos/cm	Monthly ¹
<u>Minerals</u>	mg/L	Annually ²
¹ Concurrent with effluent monitoring		
² October		

GROUNDWATER MONITORING

Groundwater monitoring is not required at this time. However, should data monitoring or other evidence indicate that waste constituents in the discharge are degrading or threatening to degrade groundwater inconsistent with the State Antidegradation Policy, this Monitoring and Reporting Program will be revised to include groundwater monitoring.

SLUDGE MONITORING

The Discharger shall collect a composite sample of sludge at least annually in accordance with EPA's *POTW SLUDGE SAMPLING AND ANALYSIS GUIDANCE DOCUMENT, AUGUST 1989*, and test for the following metals:

Arsenic	Copper	Nickel
Cadmium	Lead	Selenium
Molybdenum	Mercury	Zinc

The Discharger shall maintain sludge sampling records for a minimum of five years. A log shall be kept of sludge quantities generated, and handling and disposal activities.

The Discharger shall submit annual reports by **1 February** of each year containing the following:

- Annual sludge production in dry tons and percent solids.
- A schematic diagram showing sludge handling facilities and solids flow diagram.
- For **WWTF disposal**, include: (a) the Order numbers of WDRs that regulate the WWTF, (b) the names and locations of the facilities receiving sludge.

REPORTING

The Discharger shall report monitoring data and information as required in this Monitoring and Reporting Program and as required in the Standard Provisions and Reporting Requirements.

Monthly monitoring reports shall be submitted to the Board by the **1st day of the second month** following sample collection, and include, at a minimum, monitoring data collected during the month.

In reporting the monitoring data, the Discharger shall, in the beginning of the report, list items (if any) of noncompliance and corrective actions which have been taken or are planned to be taken to bring the discharge into full compliance. Further, the Discharger shall arrange monitoring data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly whether the discharge complies with waste discharge requirements. The Discharger shall not rely upon laboratory analytical reports to serve as tabular summaries of monitoring data. The Discharger shall maintain monitoring records for a minimum of five years.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the calculation and reporting of the values required in the discharge monitoring report. Such increased frequency shall be indicated in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to be taken to bring the discharge into full compliance with the waste discharge requirements.

By **1 February of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, titles, certificate grade (if any) and general responsibilities of persons operating and maintaining the wastewater treatment facility.
- b. The names and telephone numbers of persons to contact regarding the plant for emergency and routine situations.
- c. A certified statement of when the flow meter and other monitoring instruments were last calibrated, including identification of who did the calibration (Standard Provision C.4).
- d. A statement whether the current O&M manual and contingency plan reflect the WWTF, pond, and use area as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.

MONITORING AND REPORTING PROGRAM NO. 5-01-090
GOLDEN VALLEY UNIFIED SCHOOL DISTRICT
LIBERTY HIGH SCHOOL WWTF
MADERA COUNTY

-5-

The Discharger shall implement the above monitoring program on the first day of the month following the effective date of this Order.

Ordered by: _____
GARY M. CARLTON, Executive Officer

27 April 2001
(Date)

BLH:fmc:4/27/01

INFORMATION SHEET

ORDER NO. 5-01-079
GOLDEN VALLEY UNIFIED SCHOOL DISTRICT
LIBERTY HIGH SCHOOL WWTF
MADERA COUNTY

REPORT OF WASTE DISCHARGE

Golden Valley Unified School District (hereafter Discharger) submitted a completed Report of Waste Discharge (RWD), dated 22 May 2000, and a technical support document, dated May 2000, in support of a new wastewater treatment facility (WWTF) serving the proposed Liberty High School. Effluent will be stored in a 7-acre-foot pond and discharged to a 5-acre reclamation area. The turf area will be irrigated at agronomic rates for water and nitrogen. The WWTF and use area are on land owned by the Discharger (Assessor's Parcel No. 34-226-05/06) in Section 36, T10S, R9E, MDB&M, and in Section 33, T12S, R19E, MDB&M as shown in Attachment A. The site is north west of the intersection of Avenue 12 and Road 36, Madera County.

The school is expected to serve 675 students and staff for Phase I and 1350 students and staff for Phase II. There will be approximately 190 school days from September through early June.

The proposed WWTF will be built in two phases and consists of a package activated sludge plant with Phases I and II capacities of 20,000 and 40,000 gallons/day respectively. The WWTF includes flow monitoring, solids grinding, flow equalization, oxidation, clarification, digestion, filtration, and disinfection. The RWD states that the effluent will meet requirements for disinfected tertiary recycled water pursuant to Title 22, California Code of Regulations, Division 4, Chapter 3 (hereafter Title 22). Effluent disposal is by spray irrigation of turf at agronomic rates. During the fall and winter, effluent not required for irrigation will be stored in an unlined 7.9-acre-foot-capacity pond with a surface area of 1.35 acres. Signs will be posted around the disposal area to warn of the presence of recycled water. Sludge will be hauled off-site to a permitted publicly owned treatment works. The treatment schematic and flow diagram are shown in Attachment B, which is part of this Order by reference. The treatment flow diagram and schematic are shown in Attachment B.

A grade III operator will be required in accordance with Title 23, California Code of Regulations, Chapter 26, Section 3675 et seq.

Water Recycling Criteria

Domestic wastewater contains pathogens harmful to humans which are typically measured by means of total or fecal coliform, as indicator organisms. The California Department of Health Services (DHS), which has primary state-wide responsibility for protecting public health, has established statewide criteria in Title 22, California Code of Regulations (CCR), Section 60301 et seq., (hereafter Title 22) for the use of recycled water and has developed guidelines for specific uses.

The Discharger has filed an engineering report pursuant to Section 60323 of Title 22, which requires recyclers of treated domestic wastewater to submit an engineering report detailing the use of recycled water, contingency plans, and safeguards. The RWD states that the effluent will meet Title 22 criteria

for disinfected tertiary recycled water which may be used for irrigation of school yards. By letter dated 2 June 2000, the DHS has approved the project with recommendations for a continuous chlorine analyzer/recorder and alarm system. Those requirements are included as provisions in this permit.

Hydrology, Soils and Land Use

The proposed WWTF and use area are within the San Joaquin Valley Hydrologic Basin, San Joaquin Valley Floor Hydrologic Unit, Berenda Creek Hydrologic Area (No. 545.30) as depicted on interagency hydrologic maps prepared by the California Department of Water Resources (DWR) in August 1986. The slope is to the southwest with a grade of less than 0.1%. The site is approximately one mile north of Root Creek and four miles north of the San Joaquin River.

Site soils are predominantly Atwater loamy sand as classified by the USDA Natural Resources Conservation Service. There are small portions of Ramona sandy loam, San Joaquin sandy loams, San Joaquin-Alamo complex, Trigo fine sandy loam, Whitney fine sandy loam, and Whitney and Rocklin sandy loams. The RWD predicts 0.25 in/day percolation from the pond.

The average precipitation is 10.9 inches, according to data from the California Irrigation Management Irrigation System (CIMIS) weather station for the period of 1965 through 2000. The water balance in the RWD utilizes a 100-year return frequency annual precipitation for the project vicinity of 21.6 inches and a wet year evaporation of 60 inches.

Areal groundwater is approximately 140 to 160 feet below ground surface. Although the regional groundwater flows in the southwesterly direction, there is a localized depression such that groundwater flow is to the northwest at the project site according to information in *Lines of Equal Elevation of Water in Wells in Unconfined Aquifers*, published by DWR in Spring 1997. The RWD indicates that well yields in the project vicinity are typically 500-600 gallons/minute. According to the RWD, Liberty High School will obtain domestic and irrigation water from on-site wells.

The existing land use to the north, west, and south of the project is agricultural. The land to the east of the project is residential.

Basin Plan and Beneficial Uses

The Board adopted a *Water Quality Control Plan for the Sacramento River Basin and the San Joaquin River Basin, Fourth Edition* (hereafter Basin Plan), which designates beneficial uses, establishes water quality objectives, and contains implementation plans and policies for all waters of the basins. These requirements implement the Basin Plan. The Basin Plan identifies the present and anticipated beneficial uses of underlying groundwater as domestic, industrial, and agricultural supply.

This project complies with the reclamation objectives of the Basin Plan and the requirements for the use of recycled water for irrigation of school yards as set forth in Title 22, California Code of Regulations, Division 4, Chapter 3.

Discharge Specifications and Monitoring Requirements

The effluent limits prescribed in this Order for BOD₅, settleable solids, and EC, are based on the Basin Plan. The discharge specifications regarding dissolved oxygen and freeboard are consistent with Board policy for the prevention of nuisance conditions, and are applied to all such facilities. Title 22 requirements for irrigation of school grounds are applied since treated wastewater will be used for reclamation in the proximity of school grounds.

This Order contains influent and effluent monitoring of all constituents that required monitoring for similar orders. The addition of effluent TDS and general minerals monitoring is to develop a more accurate characterization of the discharge and its potential impact on groundwater. To determine if the Discharger is in compliance with Discharge Specification B.3.e., it is required to monitor its source water quarterly for EC and semiannually for TDS. To determine the efficiency of the Discharger's operation, the Discharger is required to monitor influent for settleable solids, BOD₅ and TSS. In order to adequately characterize its wastewater effluent, the Discharger is required to monitor continuously for chlorine residual and turbidity; daily for total coliform; weekly for pH, TSS, EC, and BOD₅; monthly for TKN, and nitrate-nitrogen; and annually for general minerals. To ensure that the pond does not create nuisance conditions, the Discharger is required to monitor freeboard available and dissolved oxygen content weekly.

Title 22, Section 60323, requires recyclers of treated municipal wastewater to submit an engineering report detailing the use of recycled water, contingency plans, and safeguards. This Order requires that biosolids be stored in a tank and to be trucked to a POTW for further treatment and disposal.

This Order requires the Discharger to collect a composite sample of sludge at least annually, in accordance with EPA's *POTW SLUDGE SAMPLING AND ANALYSIS GUIDANCE DOCUMENT, AUGUST 1989*, and test for arsenic, cadmium, molybdenum, copper, lead, mercury, nickel, selenium, and zinc. Further, this Order requires that storage, use and disposal of biosolids comply with the self-implementing federal regulations of 40 CFR 503, which are subject to enforcement by the US Environmental Protection Agency, not the Board and the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) (or any subsequent document which replaces Order No. 2000-10-DWQ).

The permitted discharge requires the Discharger to implement best practicable treatment and control for the subject wastewater and the Discharger must also ensure that the discharge does not create a condition of nuisance and maintain the highest water quality. The treatment process incorporates tertiary treatment technology. This Order requires the Discharger to submit an irrigation management plan to minimize the potential for nitrate contamination of the groundwater. This Order requires the Discharger to submit an annual report regarding sludge management and the capacity of the pond.

CEQA and Antidegradation Considerations

On 11 July 2000, the Golden Valley Unified School District Board of Trustees certified the Final Environmental Impact Report, adopted Statements and Findings of Fact, adopted a Statement of

Overriding Considerations, and adopted a Mitigation Monitoring Plan in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000 et seq.) and State CEQA guidelines.

The conditional discharge as permitted herein is consistent with the antidegradation provisions of State Water Resources Control Board Resolution No. 68-16. In the absence of groundwater quality monitoring data, it is unknown whether any assimilative capacity is available in the underlying groundwater, considering limited dilution, to allow for some degradation and not unreasonably threaten present and anticipated beneficial use of such water or result in groundwater that exceeds or threatens to exceed water quality objectives set forth in the Basin Plan. Such degradation, if it occurs, will likely be limited to groundwater underlying the pond. Some degradation of groundwater immediately underlying the pond is allowable provided the discharge reflects the implementation of BPTC measures and the degradation is: (a) consistent with the maximum benefit to the people of the State, (b) will not unreasonably affect present and anticipated beneficial use of affected groundwater, and (c) will not exceed water quality objectives.

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